

Tim Perr

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Profile

Computer Science graduate with strong experience with systems, data, and applied research. Comfortable working across the stack (React.js, Flask, SQL) and closer to the metal (C/C++, Linux). Experienced in database management, cloud computing, and machine learning. Great at problem-solving and collaborating in team environments to deliver innovative solutions. Skilled in communication and technical writing, with a track record of publishing research findings.

Education

Michigan Technological University

MS Computer Science	Aug 2024–Dec 2025
MS Data Science	Aug 2024–Dec 2025
Graduate Certificate Data Science	Aug 2024–Dec 2025
Graduate Certificate Applied Statistics	Aug 2024–Dec 2025
BS Computer Science	Aug 2021–Apr 2024

Skills

Programming Languages: Python, C/C++, Java, Swift, R, Matlab, Bash, MIPS Assembly, Makefile, LaTeX

Data and Machine Learning: SQL, Data Analysis, Machine Learning, Various Python libraries (e.g., NumPy, Pandas, Matplotlib), MATLAB Toolkits

Artificial Intelligence: LLM Creation and Analysis, Computer Vision, Fuzzy Systems, Genetic Algorithms

Web Development: React.js, Node.js, HTML, CSS, JavaScript, Flask, PHP, Rest APIs

Theoretical Knowledge: Concurrent/Parallel Computing, Quantum Computing, Data Structures, Programming Language Design, Computing Theory

Other Related Skills: Linux, Git, Agile, basic Cyber Skills, Docker

Work and Research Experience

Databases Grader

Michigan Technological University	Nov 2024 – Dec 2025
- Assisted in teaching database concepts, SQL, and database design to undergraduate students - Held office hours, graded assignments, and provided feedback to enhance student learning	

Spinel Lab Member

Michigan Technological University	Nov 2024 – Dec 2025
- Conducted research on Computer Science Education, focused in reframing perceptions of computing and developing supplemental learning materials - Performed data analysis and rewrote homework questions to enhance learning materials	

Computer Science Research Assistant

Michigan Technological University	Feb 2023 – July 2025
- Led the development team of a website for Illuminated Devices, supported by an NSF-funded grant - Presented research on the theoretical aspects of the project at ASEE 2024	

Computer Science Learning Center Coach

Michigan Technological University	Sep 2022 – Apr 2024
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- Guided students through coursework across various levels, from introductory to graduate computing courses
- Diagnosed conceptual gaps and guided students through debugging, algorithm design, and systems concepts

Instructor

iD Tech Camps	May 2022 – Aug 2022
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- Delivered curriculum to classes of up to 12 students, ensuring engagement and understanding of programming concepts
- Monitored students' progress toward learning goals and facilitated non-instructional activities

Bank Teller

First National Bank of LaGrange	Oct 2018 – Aug 2021
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- Processed cash transactions, balanced drawers and vaults, issued cashier's checks, and delivered exceptional customer service

Projects

Illuminated Devices - React.js, Flask, MariaDB, SocketIO

Objective: Create a digital portal for remote digital assistance
<https://github.com/tperr/illuminated-devices>

- Patrons go to the local library to check out device, join a waiting queue when ready, then once a Tutor is ready they are connected for assistance
- Implemented the Zoom API to connect Patrons to the Tutors
- Two publications resulted from this work

Brightness Lowerer - Python, Bash

Objective: Make a tool that lowers brightness when rapid luminosity changes occur
https://github.com/tperr/brightness_lowerer

- Grabs image of the screen, goes pixel by pixel to calculate the average luminosity
- Keeps average luminosity to a relatively consistent range
- Mostly a gimmick project, but fun and random idea

OpenFinAL - React.js

Objective: Create an open source financial analytics tool

<https://github.com/jeffreycastle/OpenFinAL>

- Programmed as coursework for BA5200 Information Systems Management with React.js, designed with Figma,
- An open source financial analytics tool designed to provide financial analytics capabilities at a low cost.
- It aims to help users understand their financial situation better and make informed decisions.
- My main focus was in the risk management module, which provides users with insights into the risk associated with their investments.